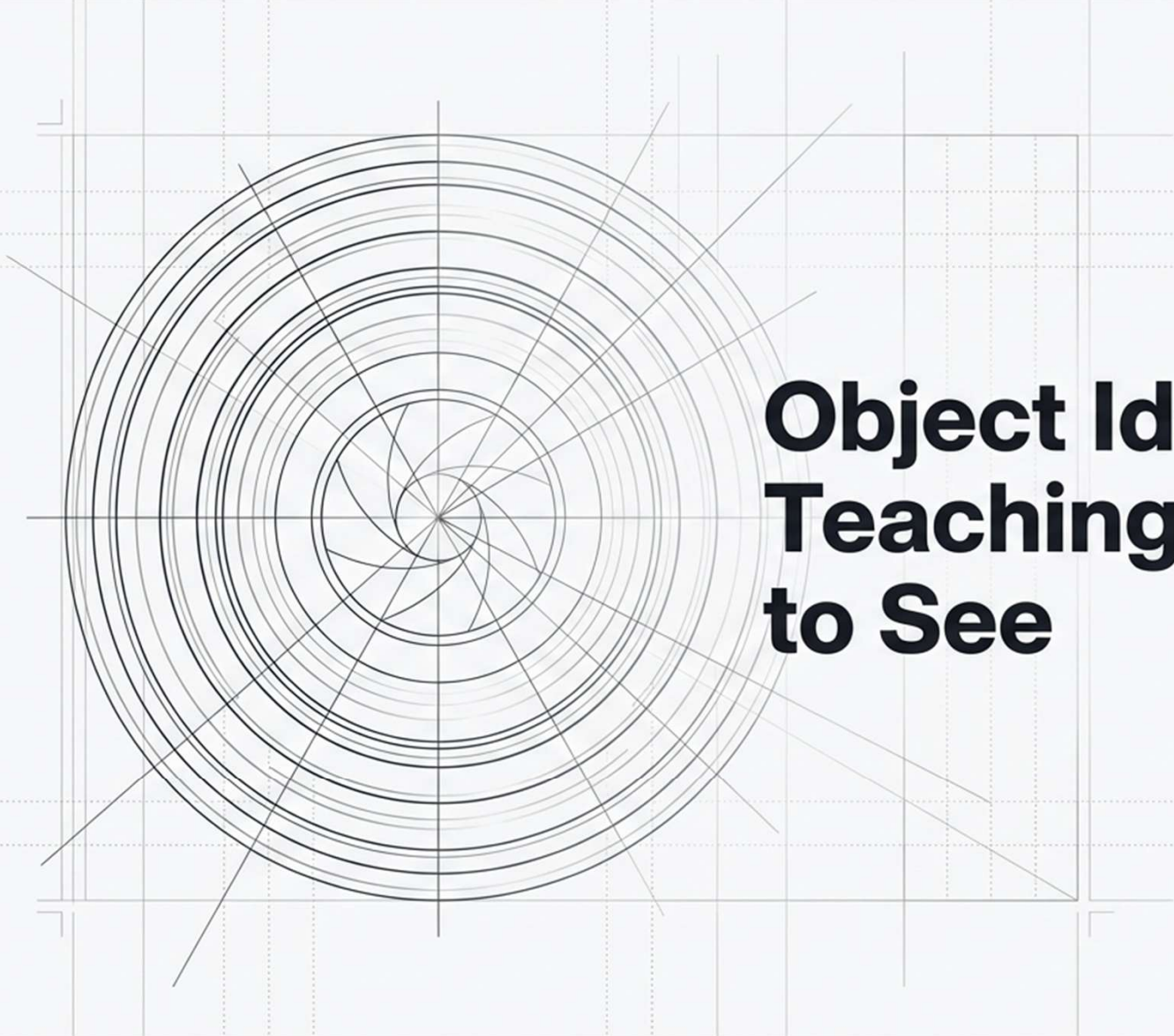
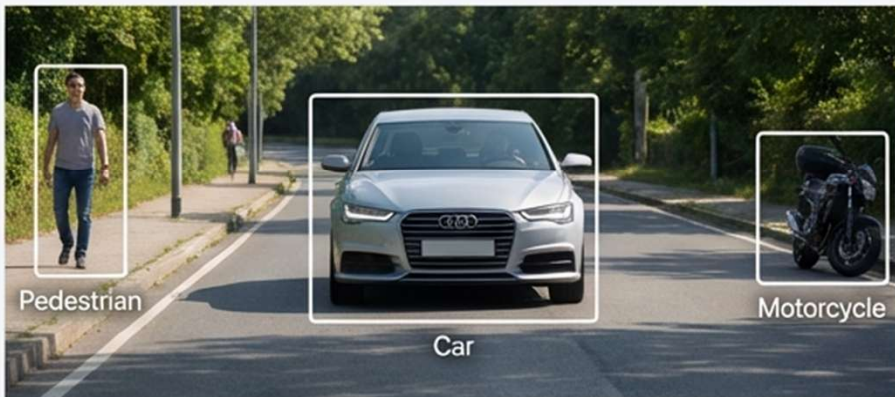




Object Identification: Teaching Computers to See

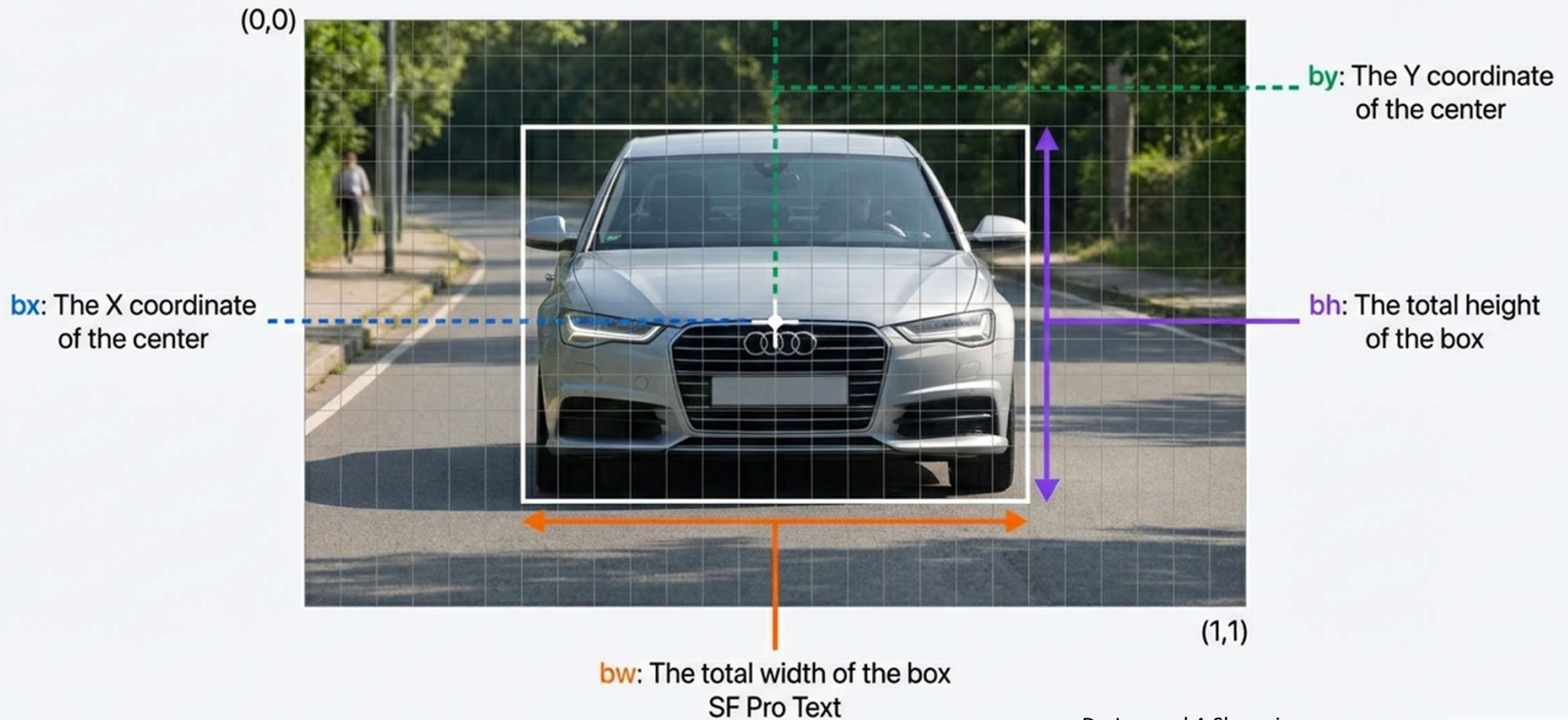
Dr. Jawwad A Shamsi
FAST-NUCES

The Four Pillars of Vision



Dr. Jawwad A Shamsi
FAST-NUCES

The Anatomy of a Bounding Box



Translating Vision to Data

$$y = \begin{bmatrix} P_c \leftarrow \text{The Master Switch: 1 if object present, 0 for background} \\ b_x \leftarrow \text{Precision Blue (\#0055FF)} \\ b_y \leftarrow \text{Emerald Green (\#00A859)} \\ b_w \leftarrow \text{Vibrant Orange (\#FF6B00)} \\ b_h \leftarrow \text{Deep Purple (\#8B3DFF)} \\ C_1 \leftarrow \text{Pedestrian} \\ C_2 \leftarrow \text{Car} \\ C_3 \leftarrow \text{Motorcycle} \end{bmatrix}$$

Example: Car Detected

[1, b_x , b_y , b_w , b_h , 0, 1, 0]

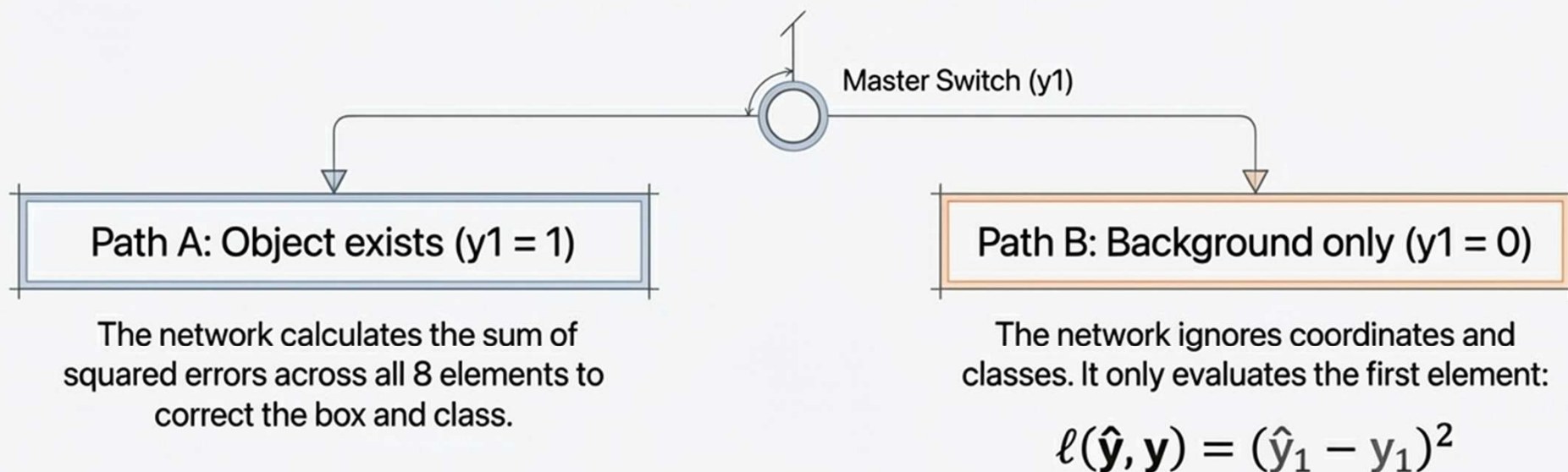
Example: Blank Background

[0, ?, ?, ?, ?, ?, ?, ?]

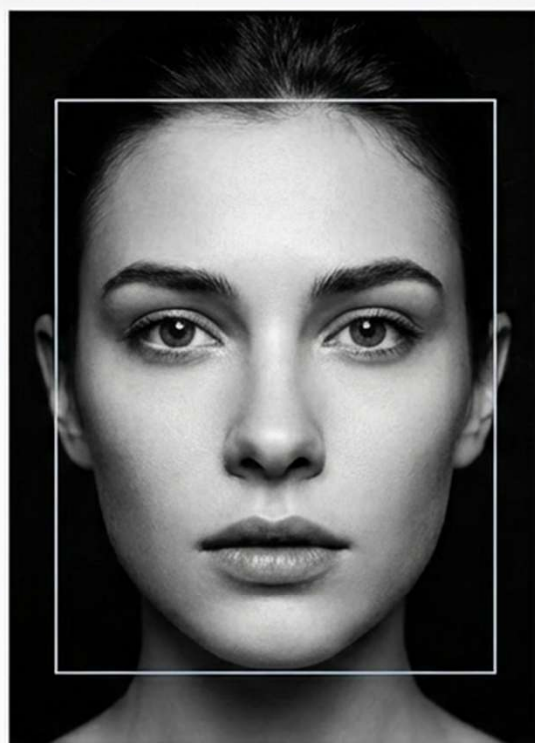
The Math of Mistakes

$$\ell(\hat{\mathbf{y}}, \mathbf{y}) = (\hat{y}_1 - y_1)^2 + (\hat{y}_2 - y_2)^2 + \dots + (\hat{y}_8 - y_8)^2$$

Log-likelihood loss can alternatively be used for C1, C2, C3.



Beyond the Box: Landmarks

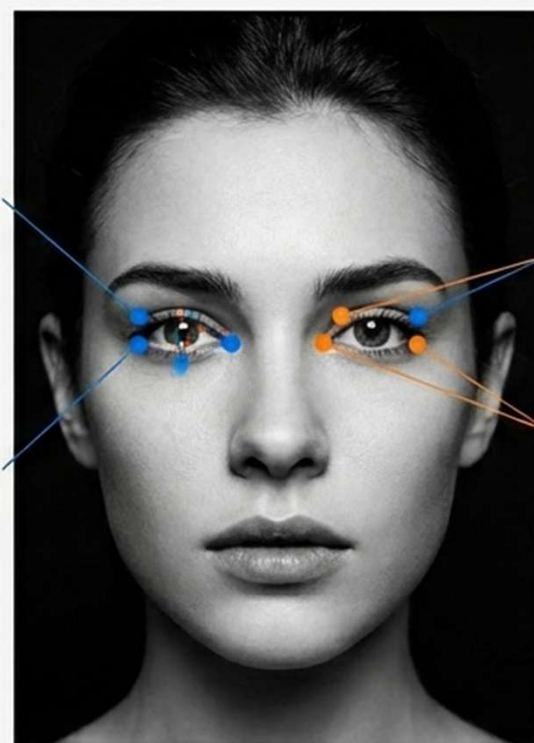


From generic regions
to granular features.



$(l1x, l1y)$

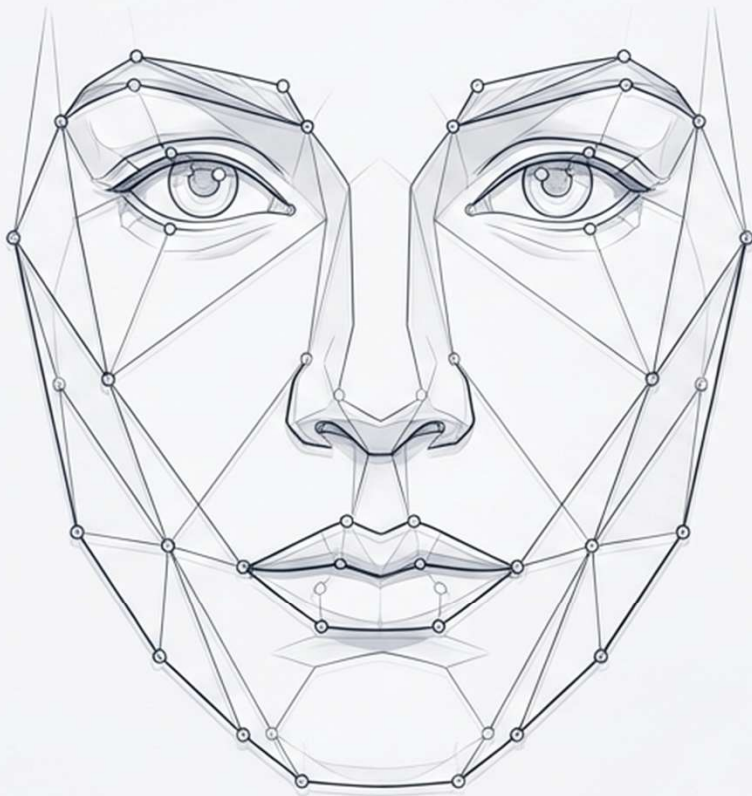
$(l2x, l2y)$



$(l3x, l3y)$

$(l4x, l4y)$

Mapping Complexity



Outputting 129 Variables:

1 variable for "Is this a face?"

64 specific (x, y) coordinate pairs

Applications requiring labeled dataset training:

- Face Recognition
- Emotion / Smile Detection
- Action Detection (e.g., Snapchat filters)

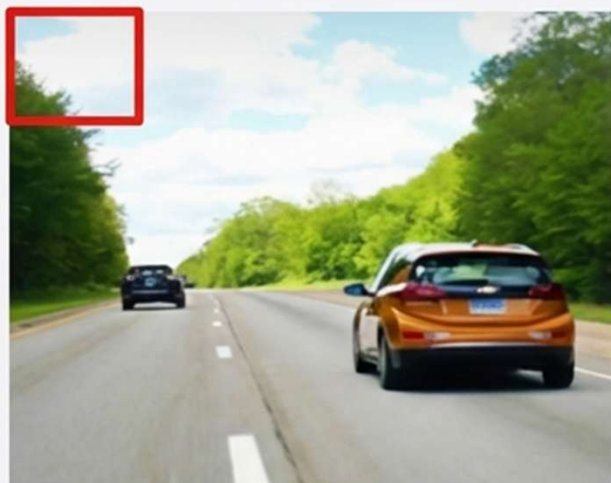
The Search Party



How do we find objects when we don't know where they are or how big they are?

Dr. Jawwad A Shamsi
FAST-NUCES

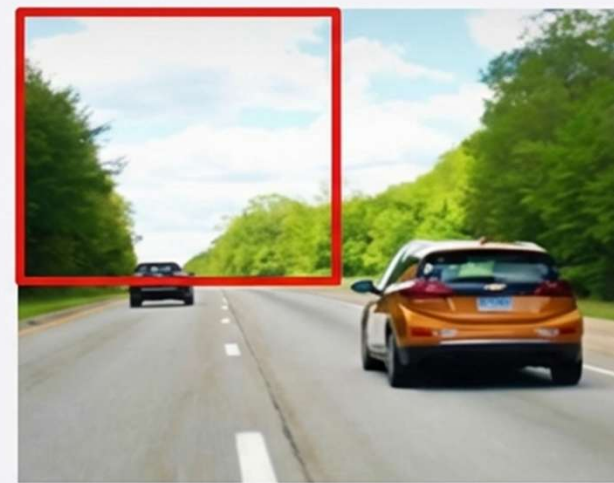
The Traditional Sliding Window



Phase 1: A small window slides across the entire image with a fixed stride.



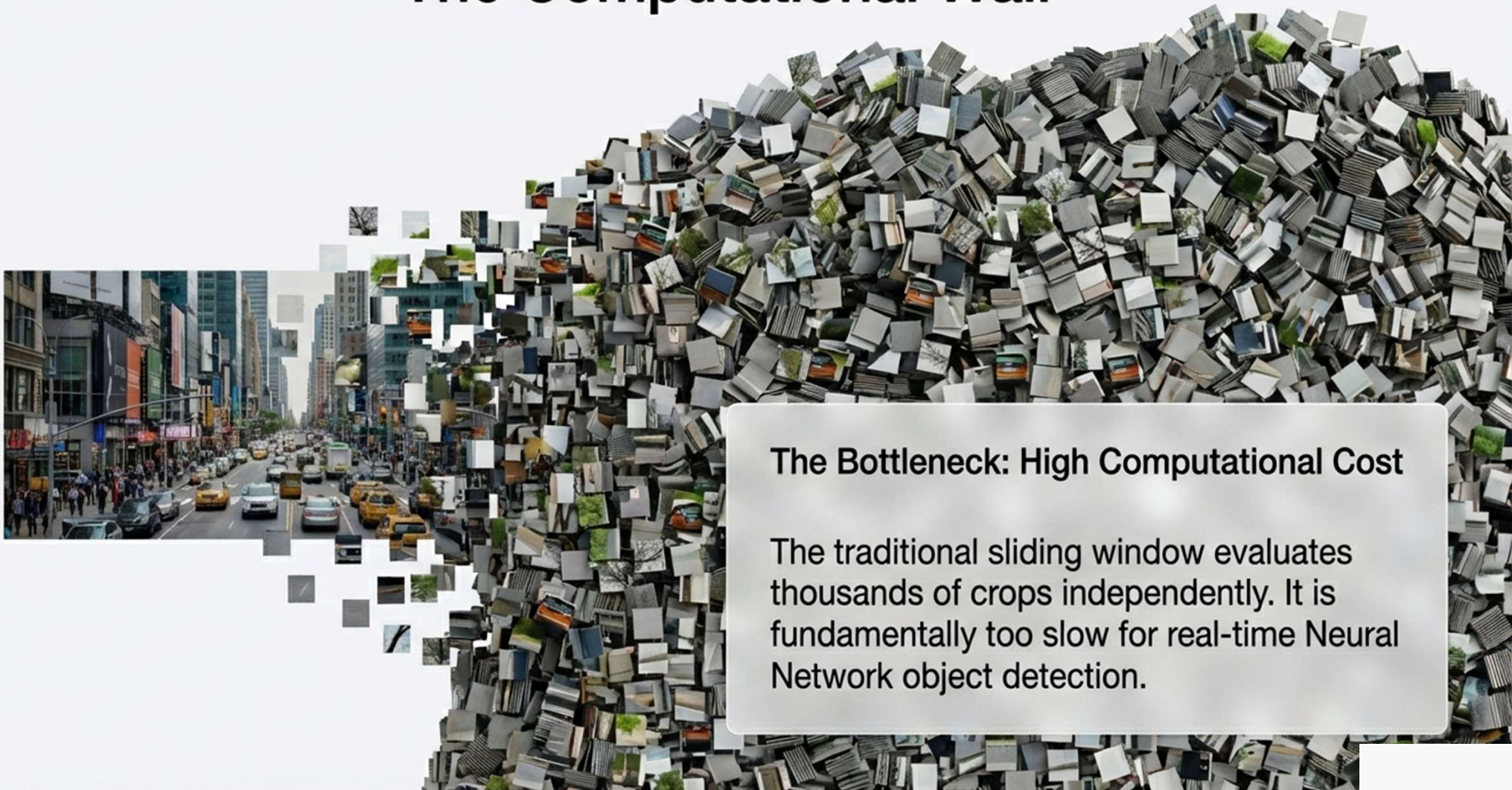
Phase 2: The window resizes (gets larger) and repeats the entire process.



Phase 3: The window resizes again to capture large objects.

A classifier is trained to evaluate every single cropped iteration individually to see if an object exists.

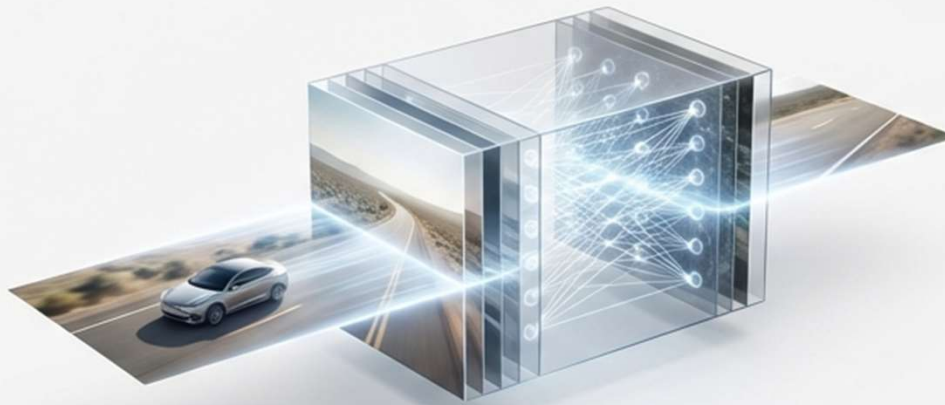
The Computational Wall



The Bottleneck: High Computational Cost

The traditional sliding window evaluates thousands of crops independently. It is fundamentally too slow for real-time Neural Network object detection.

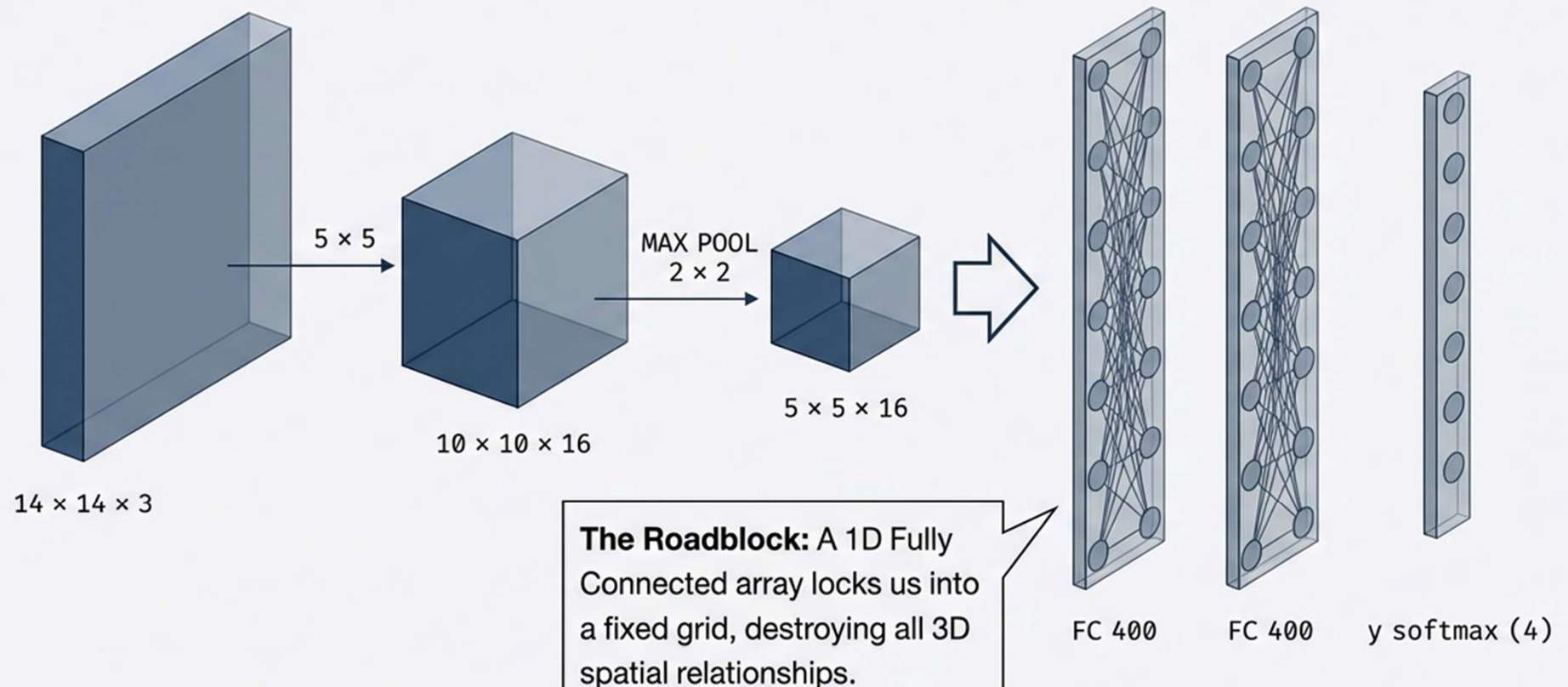
The CNN Paradigm Shift



What if we could process the entire image, and calculate every sliding window, in a single forward pass?

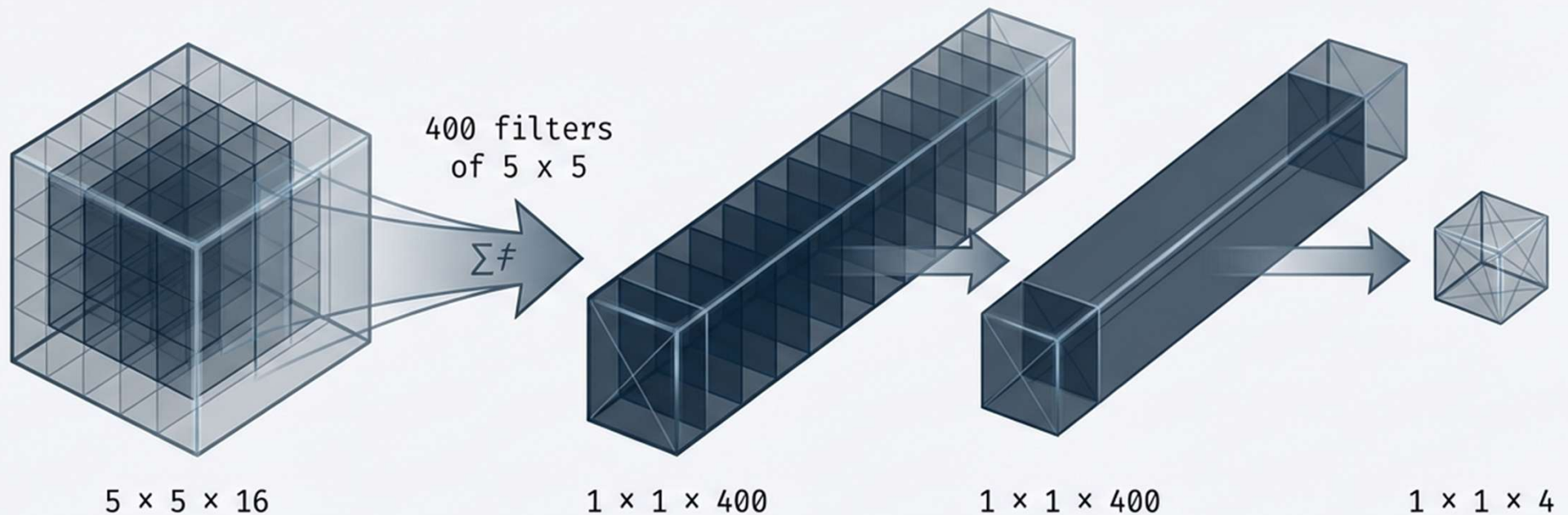
Dr. Jawwad A Shamsi
FAST-NUCES

The Architecture Problem



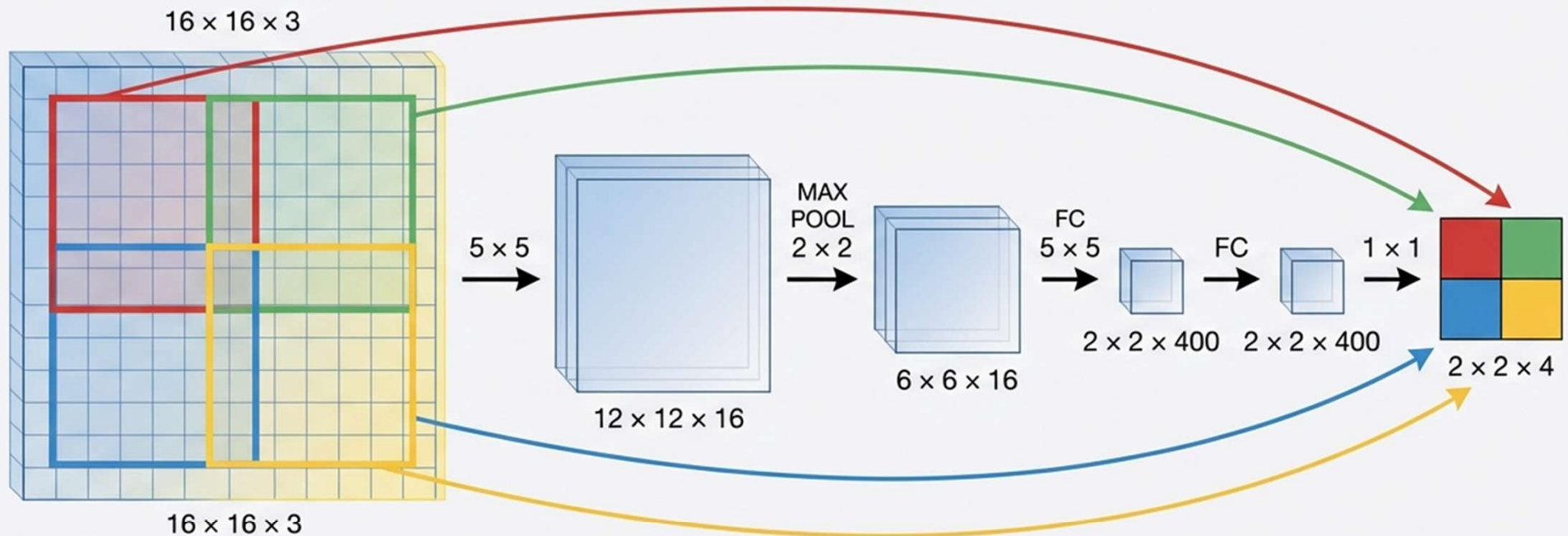
Dr. Jawwad A Shamsi
FAST-NUCES

The Convolutional Transformation



The Solution: We execute identical mathematical operations, but convert the 1D FC array into a spatial Convolutional layer. The 3D architecture of the image is preserved.

Sliding Windows Without the Slide



Each cell in the 2×2 output mathematically corresponds to a discrete 14×14 sliding window crop. The network computed all four windows simultaneously in a single pass.

Real-Time Vision Achieved

1. **Identify (Target Vector y)**
The 8-variable array translating physical objects into binary data.
2. **Locate (Bounding Boxes & Landmarks)**
From mapping 4 core coordinates to extracting 129 facial landmarks.
3. **Execute (Convolutional Windows)**
Replacing agonizing pixel searches with a single mathematical pass.

